ETH zürich

Plotting Wavefields from Research Module

Salome Bachmann

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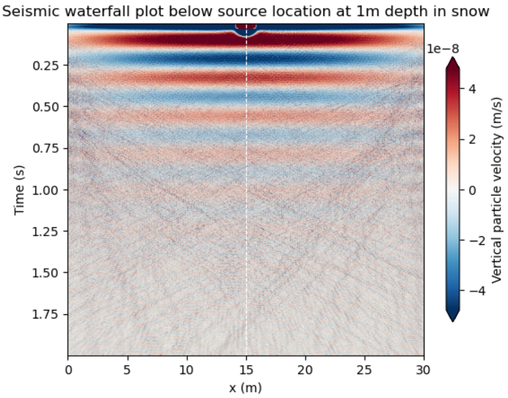
What have I been working on?

- Writing introduction
- Implementing feedback for introdcution
- Setup of remote machine
- Running simulations with lower source frequency
- Plotting waterfall plots, wiggle plots, single traces and frequency content

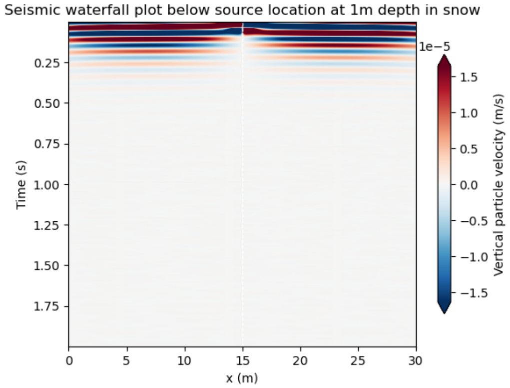
Simulation parameters used

- 30m by 3m domain
- one simulation with 150m by 3m domain
- 3 layers, all layers 1m thick
- absorbing boundaries at edges of domain
- 10Hz vector point source
- moment tensor source for skier (myy=0, mxx=0, mxy=3e4)

Wavefield Waterfall plots

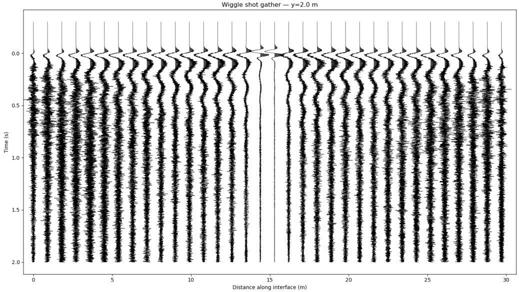


(a) Vector point 2D source with $f_y = -1$. Source location indicated by white line. Colorbar shows particle velocity (m/s).

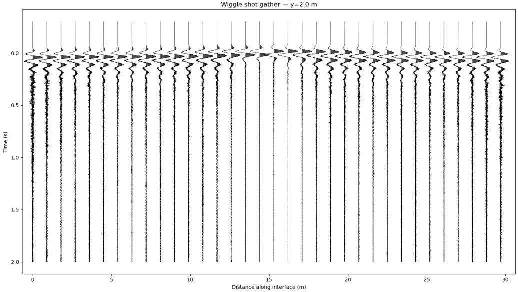


(b) Moment tensor source with $m_{xx} = 0$, $m_{yy} = 0$, $m_{xy} = 3 \times 10^4$. Source location indicated by white line. Colorbar shows particle velocity (m/s).

Wiggle Plots

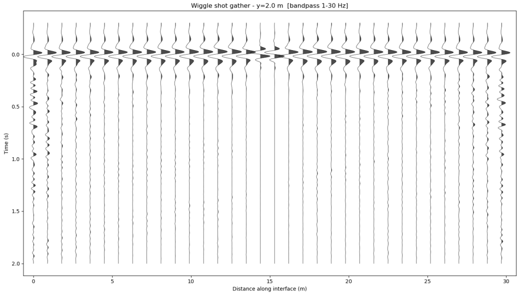


(a) Wiggle plot for vector point source.

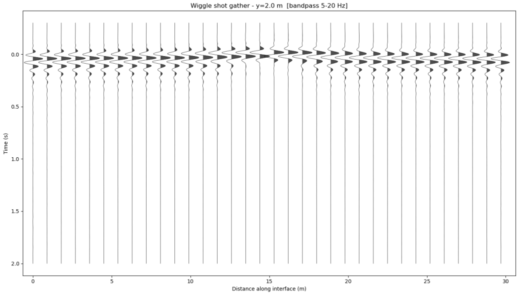


(b) Wiggle plot for moment tensor source.

Bandpass filtered wiggle plots



(a) Bandpass filtered wiggle plot for vector point source, Bandpass from 1 to 30HZ



(b) Bandpass filtered wiggle plot for moment tensor source, Bandpass from 5-20HZ

Single traces

Problems / Questions

- Simulations take very long to run on remote machine (2+ hours)
- Does it make sense to add topography to simulations?
- Does it make sense to add eg old snow layer (make the simulations multilayered)?

Salome Bachmann
Master Student
sbachmann@student.ethz.ch

ETH Zurich
D-EAPS